

ASSIGNMENT - 2

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Code : C1 extends C2

```
// create an class
class c1{
    int a = 24;
    void sos(){
        System.out.println("this is my first class");
    }
}

// class 1 inherits the second class by using extends keyword
public class Assign2 extends c1 {
    // to modify the method in super class
    @Override
    void sos(){
        System.out.println("the value of a is" + a);
        System.out.println("this is my first class");
    }

    public static void main(String[] args) {
        // we create an object for sub class
        Assign2 ab = new Assign2();
        ab.sos();
    }
}
```

result :

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Assign2.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Assign2
the value of a is24
this is my first class
```


Second question:

C1 extends C2 C3

```
// we create the class of three classes
class animal{
    String name = " lion";
    void ani(){
        System.out.println(" lion is the my fav animal");
    }
}

// super class is in subclass by using keyword called extends
class bird extends animal{
    String names = " parrot , sparrow";
    void bi(){
        System.out.println("parrot is my fav bird");// multi inheritance is executed
    }
}

// super class is in subclass by extends keyword
class sea extends animal {
    float marks = 99.23f;
    void mar(){
        System.out.println(" you got 1 st rank");
    }
}

public class Assign21 {
    public static void main(String[] args) {
        // we create an object of 3rd class only
        sea as = new sea();
        as.mar();
        as.ani();
        as.ani();

    }
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Assign21.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Assign21
you got 1 st rank
lion is the my fav animal
lion is the my fav animal
```

C1 implements I1

// we can create the an interface for that class

```
interface one{  
    int a =23;  
    void two();  
}
```

// the class is innerited the interface

```
public class Assign22 implements one {  
    public void two(){
```

```
System.out.println("the value of interface is"  
    +a);  
    }
```

```
public static void main(String[] args) {
```

```
    //we create the object
```

```
    Assign22 ac = new Assign22();
```

```
        ac.two();
```

```
    }
```

```
}
```



```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Assign22.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Assign22
the value of interface is23
```

C1 implements I1 and I2

```
interface first{
    int a =244;
    void system();
}

// in second interface have an initialised variable
// and abstract method called part
interface second{
    int y = 300;
    void part();
}

// to inherit the interfaCes by using implements
// keyword
public class Assign23 implements first ,second {
    // to override the methods
    @Override
    public void system(){
        System.out.println("the multiple inheritance
        value is" + a);
    }
    @Override
    public void part(){
        System.out.println("the second interface of
        value is " + y);
    }

    public static void main(String[] args) {
        Assign23 as = new Assign23();
        as.system();
        as.part();
    }
}
```



```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Assign23.java
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Assign23
the multiple inheritance value is 244
the second interface of value is 300
```

// two interfaces are inherited by one class by
using implements keyword

```
interface one{
    int a =244;
    void at();
}
interface two extends one{
    int q= 4;
    void ta();
}

class many implements one,two{
    @Override
    public void at(){
        System.out.println("value is" +a);
    }
    @Override
    public void ta(){
        System.out.println("percent is" +q);
    }
}

public class Interface1 {
public static void main(String[] args) {
    // we create an object for class
    many m = new many();
    System.out.println(m.a);
    m.at();
    System.out.println();
    m.ta();

}
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Interface1.java
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Interface1
```

```
244
```

```
value is244
```

```
percent is4
```


// multi interfaces are inherited are one class and
divide into last interface are implemented by d
interface

```
interface a{
    int x = 44;
    void it();
}
interface b{
String y = "hemanth";
    void ti();
}

interface c {
    double z = 44.98;
    void ui();
}
interface d extends a,b,c{
    int b = 99;
    void iu();
}
class many implements d{
    @Override
    public void it(){
        System.out.println(x);
    };
    @Override
    public void ti(){
        System.out.println(y);
    };
    @Override
    public void ui(){
        System.out.println(z);
    };
    @Override
    public void iu(){
        System.out.println(b);
    };
}

public class Interface2 {
    public static void main(String[] args) {
// we can create an object for Many class
        many n = new many();
        System.out.println(n.y);
        System.out.println(n.x);
        System.out.println(n.z);
        System.out.println(n.b);
        n.it();
        n.iu();
        n.ti();
        n.ui();
    }
}
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Interface2.java
```

```
PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Interface2
```

```
hemanth
```

```
44
```

```
44.98
```

```
99
```

```
44
```

```
99
```

```
hemanth
```

```
44.98
```

```
// one class is inherited super class and interface
also
// class to class extends
// class to interface implements
```

```
class t{
    int y = 44;
    t(){
        System.out.println(y);
    }
    void tv(){
        System.out.println(y+9);
    }
}
```

```
interface v {
    int z =99;
    void rt();
}
```

```
class u extends t implements v{
    int x= 24;
    @Override
    void tv(){
        System.out.println(y+200);
    }
    u(){
        System.out.println(x);
    }
    public void rt(){
        System.out.println(x+20);
    }
}
```

```
public class Interface3 {
    public static void main(String[] args) {
```

```
        u p = new u();
        p.tv();
        p.rt();
    }
}
```


44.98

PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> **javac** Interface3.java

PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> **java** Interface3

44

24

244

44

//!one class is taking by inherited of one super class and multiple
interfaces

// class to class and extends

//class to interfaces implements

```
class t{
    int y = 44;
    t(){
        System.out.println(y);
    }
    void tv(){
        System.out.println(y+9);
    }
}

interface v {
    int z =99;
    void rt();
}

interface w{
    void ov();
}

class u extends t implements v ,w{
    int x= 24;
    @Override
    void tv(){
        System.out.println(y+200);
    }
    u(){
        System.out.println(x);
    }
    public void rt(){
        System.out.println(x+20);
    }
    public void ov(){
        System.out.println("multiple inheritance is completed");
    }
}

public class Interface4 {
    public static void main(String[] args) {

        u o = new u();
        o.ov();
        o.rt();
        o.tv();

    }
}
```


44

PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> javac Interface4.java

PS C:\Users\bsai8\OneDrive\Desktop\javaprograms> java Interface4

44

24

multiple inheritance is completed

44

244